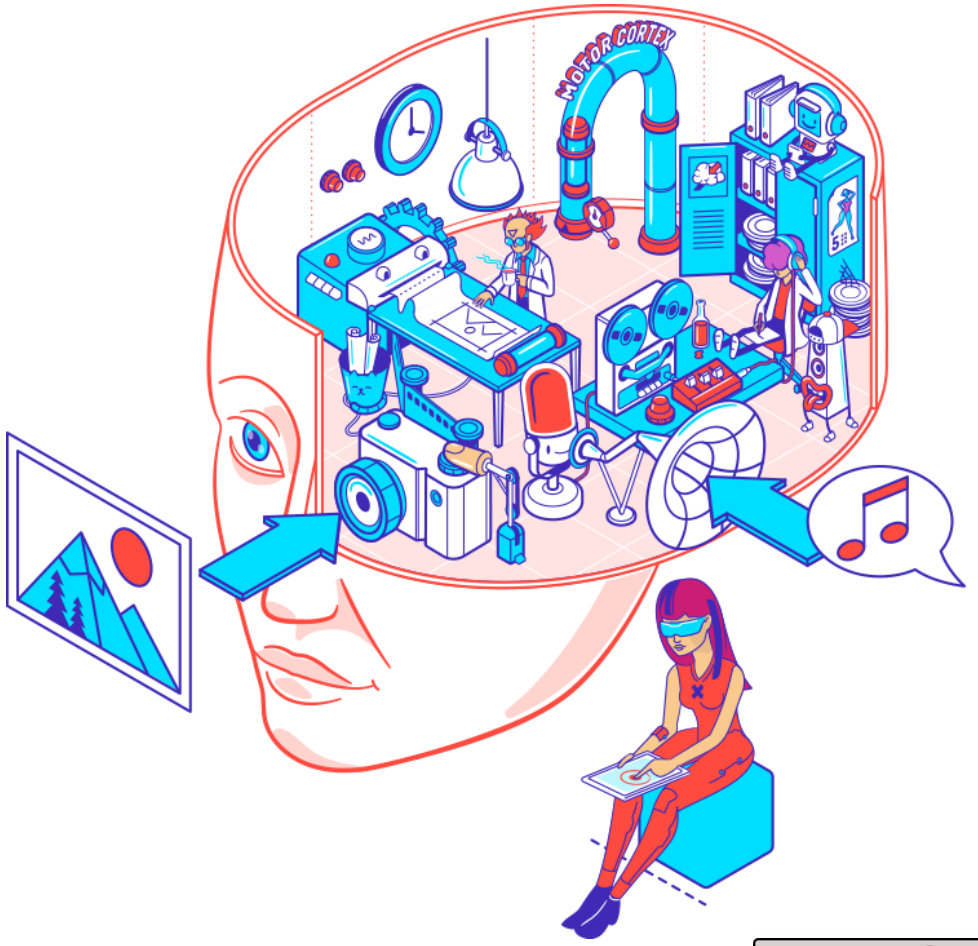
HAW HAMBURG

MEDIA PROGRAMMING

UNIT X: CLOSING



Prof. Dr. Eike Langbehn

Department of Media Technology
Faculty of Design, Media and Information
Hamburg University of Applied Sciences

CC BY

LEARNING OUTCOME

- Die Studierenden lernen die Grundlagen digitaler Medien und wenden programmiertechnische Methoden an um digitale Medien zu **verarbeiten**, zu **speichern**, zu **manipulieren** und zu **erstellen**.

- Die Studierenden
  • kennen verbreitete Formate digitaler Medien
  • können grundlegende Algorithmen zur Verarbeitung und Manipulation von digitalen Medien implementieren

LEARNING OUTCOME

- Erwerb von grundlegenden Kenntnissen für die Entwicklung von Grafik-basierten Anwendungen.
- Die Studierenden können Methoden anwenden um interaktive Inhalte digitaler Medien zu erstellen.

RECAP

UNIT X: CLOSING

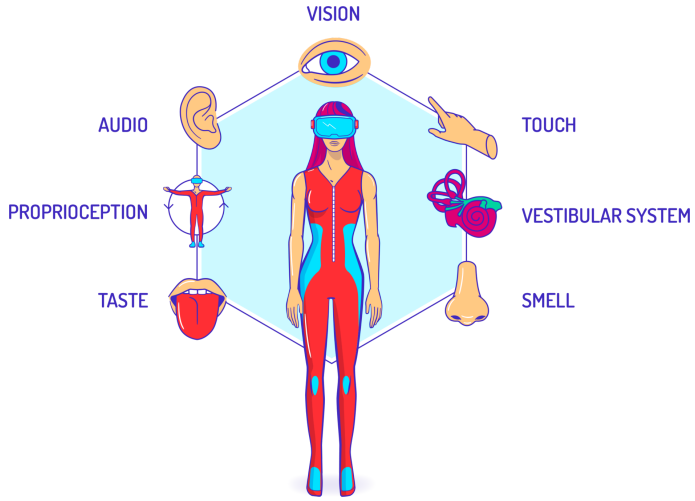


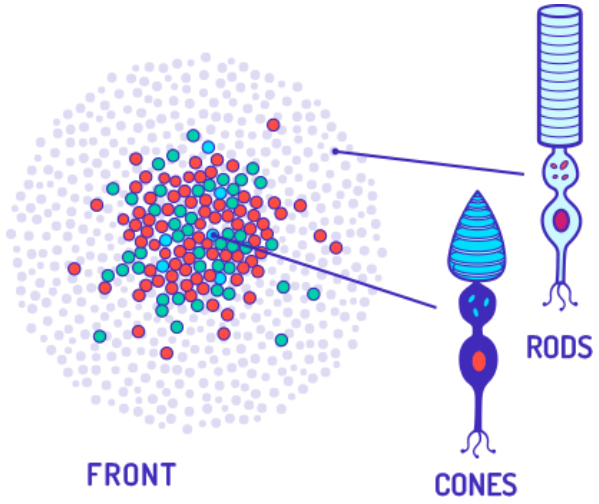
AGENDA:

1. Introduction / Human Perception
2. From Java to Python
3. Raster Graphics
4. Algorithmic Graphics Generation
5. Vector Graphics
6. Interactive Graphics
7. Animated Graphics

1. HUMAN PERCEPTION

- Sensory Modalities
- Visual Perception
- Human Eye





3. RASTER GRAPHICS

- Advantages and Disadvantages
- Encoding
- JPEG Compression
- Point Ops and Filter

4. GRAPHICS GENERATION

- Function-based Image Synthesis (Gradients)
- Random Numbers
- Noise (Perlin Noise)
- Fractals (Mandelbrot-Set)

5. VECTOR GRAPHICS

- Advantages and Disadvantages
- Vectors, Primitives and Curves
- Transformations (Matrices)
- Encoding (SVG)
- Rendering (Rasterization)

6. INTERACTIVE GRAPHICS

- Main Loop
- Delta Time
- GDScript

7. ANIMATED GRAPHICS

- Frame-by-Frame Animations (Spritesheets)
- Key-Frame Animations
- Procedural Animations (Tweening, sin/cos)

OUTLOOK

UNIT X: CLOSING

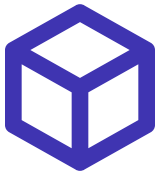


OUTLOOK



COMPUTER VISION

Image processing, feature detection, and scene understanding



COMPUTER GRAPHICS

Advanced rendering, shading, and animation in 3D



GAME DEVELOPMENT

Interaction, physics engines, and real-time optimization



Computer Vision

3D-Modellierung und Animation



Game Design / Integrationsprojekt: Games

EXAM

UNIT X: CLOSING



EXAM

You should look again at all content:

- Slides and Videos
- Quizzes
- Exercises

EXAM

- Not necessary to write syntax-correct python code
- Write pseudo-code
- Understand what given code does
- Fill in gaps in given code

QUESTIONS?

UNIT X: CLOSING



HAW HAMBURG

Unless otherwise noted, the contents of this Open Educational Resource are licensed under the Creative-Commons Attribution 4.0 International license (CC BY 4.0)
Attribution: Rafael Epplée, Eike Langbehn, Peter Wood

